How do we convert a 2nd order ordinary differential equation (ODE) to a 2 x 2 system of differential equations (SDE)? More generally, how do we convert an n-th order ODS to an n x n SDE?

Answer:

Converting a 2nd Order ODE to a 2x2 SDE

The Equation:

Suppose you have a second-order linear ODE:

$y'' + ay' + by = 0$

Step 1: Define New Variables

Let's create two variables for the SDE:

- $y_1 = y$ (1)
- $y_2 = y'$ (2)

Step 2: Differentiate the New Variables in (1) and (2)

Now, find the derivative of each:

- $y_1' = y'$ (3)
- $y_2' = y''$ (4)

Step 3: Substitute the Original Equation

Replace y'' in (4) using your original ODE (solved for the highest derivative):

$y_2' = -ay_2 - by_1$ (5)

The Resulting 2x2 System:

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= -by_1 - ay_2 \end{aligned}$$

2. General Case: n-th Order ODE to n x n SDE

To convert an n-th order ODE, you simply extend this logic by creating n variables, each representing a successively higher derivative.

The Equation:

$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = 0$

The Substitution:

Define variables $y_1, y_2, \dots, y_n$:

- $y_1 = y$
- $y_2 = y'$
- $y_3 = y''$
- ...
- $y_n = y^{(n-1)}$

The First-Order System:

This leads to a chain of simple relations, with the final equation capturing the original ODE's dynamics:

$$
\begin{aligned}
y_1' &= y_2 \\
y_2' &= y_3 \\
&\vdots \\
y_{n-1}' &= y_n \\
y_n' &= -a_0 y_1 - a_1 y_2 - \cdots - a_{n-1} y_n
\end{aligned}
$$

Matrix Form

In linear algebra, we write this general system in matrix form, **y' = Ay** :

$$
\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}
$$